

# Answers learning task 12

2026 – 2027

**Exercise 7.34:** Let  $X$  be a geometric random variable with parameter  $p$ . Find the maximum likelihood estimator of  $p$  based on a random sample of size  $n$ .

Answer

$$\hat{p} = \frac{1}{\bar{X}}$$

**Exercise 7.35:** Consider the Poisson distribution with parameter  $\lambda$ . Find the maximum likelihood estimator of  $\lambda$ , based on a random sample of size  $n$ .

Answer

$$\hat{\lambda} = \bar{X}$$

**Exercise 7.38:** Consider the probability density function

$$f(x) = \frac{1}{\theta^2} x e^{-x/\theta}, \quad 0 \leq x < \infty, \quad 0 < \theta < \infty$$

Find the maximum likelihood estimator for  $\theta$ .

Answer

$$\hat{\theta} = \frac{\bar{X}}{2}$$

**Exercise 7.40:** Consider the probability density function

$$f(x) = c(1 + \theta x), \quad -1 \leq x \leq 1$$

- (a) Find the value of the constant  $c$ .
- (b) Show that  $\hat{\theta} = 3\bar{X}$  is an unbiased estimator for  $\theta$ .

Answer

- (a)  $c = \frac{1}{2}$
- (b)  $E(\bar{X}) = E(X) = \frac{1}{3}\theta$ . So  $E(3\bar{X}) = \theta$ .

**Exercise 7.41:** The Rayleigh distribution has probability density function

$$f(x) = \frac{x}{\theta} e^{-x^2/2\theta}, \quad x > 0, \quad 0 < \theta < \infty$$

- (a) It can be shown that  $E(X^2) = 2\theta$ . Use this information to construct an unbiased estimator for  $\theta$ .

- (b) Find the maximum likelihood estimator of  $\theta$ . Compare your answer to part (a).

### Answer

- (a)  $\hat{\theta} = \frac{\sum_{i=1}^n X_i^2}{2n}$   
 (b)  $\hat{\theta} = \frac{\sum_{i=1}^n X_i^2}{2n}$ , the maximum likelihood estimator is identical to the unbiased estimator in part (a).

**Exercise 7.42:** Let  $X_1, X_2, \dots, X_n$  be uniformly distributed on the interval 0 to  $a$ . Recall that the maximum likelihood estimator of  $a$  is  $\hat{a} = \max(X_i)$ .

- (a) Argue intuitively why  $\hat{a}$  cannot be an unbiased estimator for  $a$ .  
 (b) Suppose that  $E(\hat{a}) = na/(n+1)$ . Is it reasonable that  $\hat{a}$  consistently underestimates  $a$ ? Show that the bias in the estimator approaches zero as  $n$  gets large.  
 (c) Propose an unbiased estimator for  $a$ .  
 (d) Let  $Y = \max(X_i)$ . Use the fact that  $Y \leq y$  if and only if each  $X_i \leq y$  to derive the cumulative distribution function of  $Y$ . Then show that the probability density function of  $Y$  is

$$f(y) = \begin{cases} \frac{ny^{n-1}}{a^n}, & 0 \leq y \leq a \\ 0, & \text{otherwise} \end{cases}$$

Use this result to show that the maximum likelihood estimator for  $a$  is biased.

- (e) We have two unbiased estimator for  $a$ : the moment estimator  $\hat{a}_1 = 2\bar{X}$  and  $\hat{a}_2 = [(n+1)/n] \max(X_i)$ , where  $\max(X_i)$  is the largest observation in a random sample of size  $n$ . It can be shown that  $V(\hat{a}_1) = a^2/(3n)$  and that  $V(\hat{a}_2) = a^2/[n(n+2)]$ . Show that if  $n > 1$ ,  $\hat{a}_2$  is a better estimator than  $\hat{a}_1$ . In what sense is it a better estimator of  $a$ ?

### Answer

- (a) The largest observation is almost always less than  $a$ .  
 (b) Since  $\frac{n}{n+1} < 1$ , it consistently underestimates  $a$ . The bias is  $\frac{na}{n+1} - a = -\frac{a}{n+1} \rightarrow 0$  as  $n \rightarrow \infty$ .  
 (c)  $\hat{a} = \frac{n+1}{n} X_{\max}$   
 (d)  $F(y) = \left(\frac{y}{a}\right)^n \Rightarrow f(y) = \frac{ny^{n-1}}{a^n} \Rightarrow E(Y) = \frac{na}{n+1} \Leftrightarrow E\left(\frac{n+1}{n}Y\right) = a$ , showing that  $Y = \hat{a}_{\text{MLE}}$  is biased.  
 (e) Since  $n(n+2) > 3n \Leftrightarrow n < 0$  or  $n > 1 \Leftrightarrow n > 1$ ,  $V(\hat{a}_2) < V(\hat{a}_1)$  for  $n > 1$ . Thus,  $\hat{a}_2$  is a better (i.e., more precise) estimator than  $\hat{a}_1$ .